

Data Lake

Repository for large volumes of structured and unstructured data used for analytics and processing.

Concept

Data Lakes store varied data formats, often before final analytical modeling.

Official site and documentation

Use the official Data Lake source as the primary reference: <https://aws.amazon.com/what-is/data-lake/>.

Official documentation should be the first source for installation, supported versions, syntax changes and new features.

Risks

Schema drift, duplicates, incorrect partitions, permissions, late data and ingestion quality are common risks.

QA

Validate ingestion, metadata, partitions, volume, transformation rules and access.

Lake vs Warehouse

Data Lakes favor flexibility/raw data; Data Warehouses usually organize modeled data for analytics consumption.

When to use it

Use this knowledge when product risk, architecture or the delivery stage requires this type of validation. Confirm the goal, environment, required data and expected evidence before starting.

How QA applies it

Start from the behavior or risk to validate, prepare data and environment, execute the technique or tool reproducibly and capture evidence that helps the team understand the result.

What it is and where it fits

Data Lake is a tool or technology that can support Quality Engineering work. Before adopting it, understand the problem it solves, where it fits in the workflow and what evidence or productivity gain it should provide.

First QA-oriented use

Pick one small, verifiable task in Data Lake, configure only what is necessary, run a simple example and confirm the outcome. Repeat with different data or conditions before adding abstractions.

Using it in a real project

Define where Data Lake fits, who owns configuration, how credentials and data are handled and how results are shared. Integrate it into the team workflow instead of creating an isolated activity.

Adoption good practices

Standardize versions and configuration, document important decisions, keep secrets out of code and preserve reproducible examples. Reassess whether Data Lake still delivers value as the product evolves.

Common issues and troubleshooting

When Data Lake fails, verify versions, dependencies, permissions, connectivity, data and environment before assuming a product defect. Preserve the original error and reduce the issue to the smallest reproducible case.